EQUAL Platform: Complete Software Requirements Specifications (SRS)

Software Requirements Specification (SRS)

AI Prompts & Change Points Configuration System

Document Control

Item	Details
Document Title	SRS: Pre-Upload Column Report & Editable AI Prompts with Change Points
System Feature	Pre-Upload AI Pipeline Modal (`AiPipelinePromptModal`)
Core AI Engine	GPT Opus 120B Large Language Model
Document Version	1.0.0 (Pre-Implementation Baseline)
Target Audience	Domain Experts, System Administrators, QC Engineers

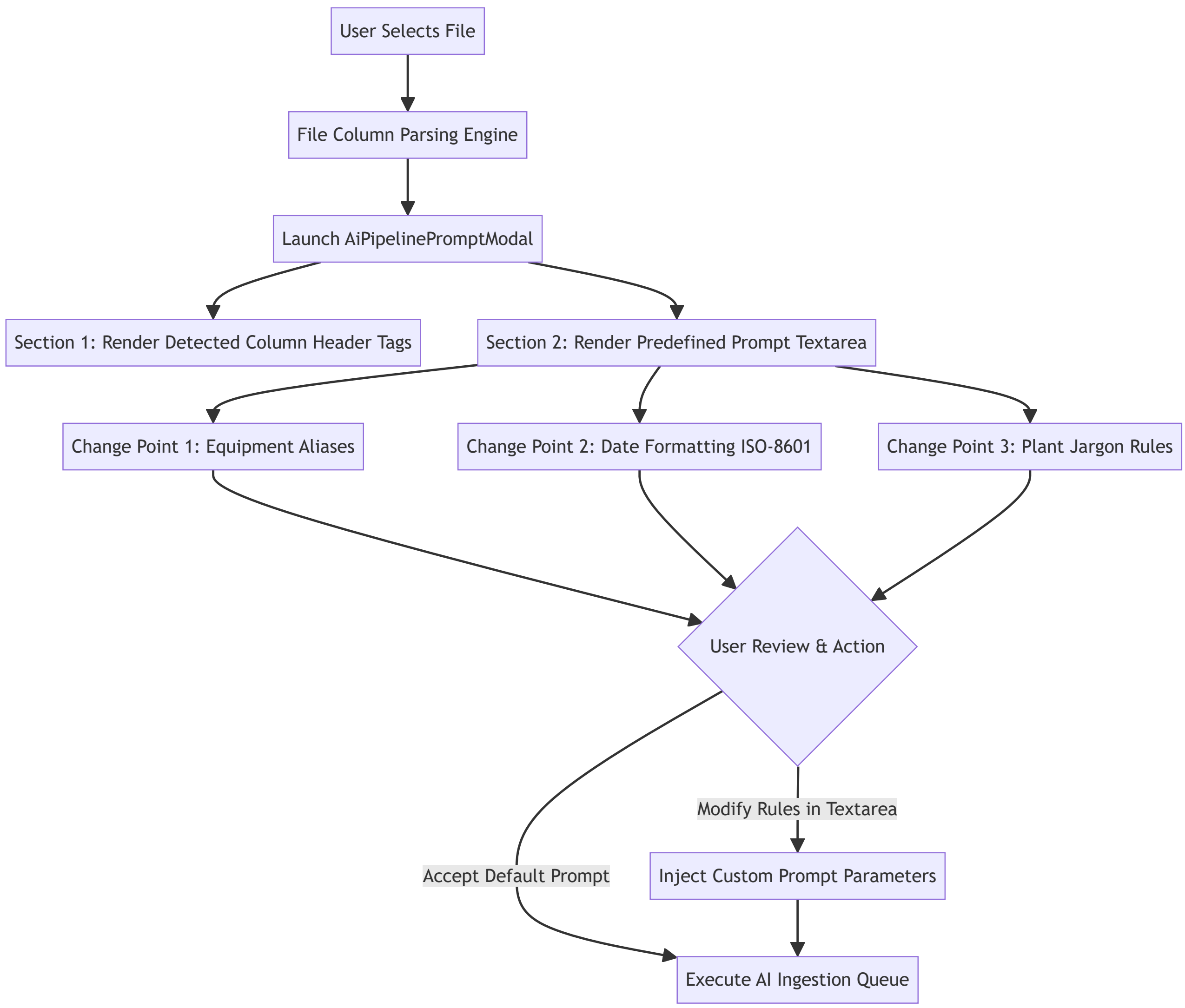
1. Functional Scope & Individual Flow Diagram

1.1 Objective

When uploading datasets through the AI Pipeline, operators MUST be presented with a **Pre-Upload Column Summary Report** displaying all detected input columns. Immediately below the report, the system MUST display a **Predefined System Prompt** configured with defined Change Points. Operators can review and customize extraction rules prior to executing pipeline processing.

1.2 Individual Module Flow Diagram

SYSTEM WORKFLOW & PROCESS ARCHITECTURE



2. Component Requirements

2.1 Detected Column Summary Report Page

- **Requirement**: Displays an interactive summary of all column headers parsed from the selected input file.
- **QC Verification**: Ensure column count and header labels match source file headers exactly.

2.2 Predefined AI Prompt & Change Points Component

- **Requirement**: Displays an editable prompt template containing domain rules for the **GPT Opus 120B** model:
- **Change Point 1: Equipment Alias Normalization** (e.g., mapping "R203" to "0303. R2 Coater").
- **Change Point 2: Date Formatting Directive** (enforcing ISO-8601 `YYYY-MM-DD`).
- **Change Point 3: Jargon Translation Directive** (standardizing plant abbreviations).
- **User Control**: The prompt box MUST be fully editable by the user before initiating extraction.

3. QC Testing & Acceptance Criteria

- **Report Display Verification**: Modal MUST display detected columns accurately before pipeline execution.
- **Editable Prompt Verification**: User modifications to the prompt text MUST be passed into the extraction run.
- **Change Point Effect Verification**: QC testers MUST confirm that custom rules added to the prompt take effect during data extraction.

Software Requirements Specification (SRS)

Change History Data Management Module

Document Control

Item	Details
Document Title	SRS: Change History Data Management Workspace
System Module	EQUAL Data Management Subsystem (`/data-management/change-history-data`)
Document Version	1.0.0 (Pre-Implementation Baseline)
Target Audience	Software Engineers, QA / QC Test Engineers, Enterprise Solution Architects

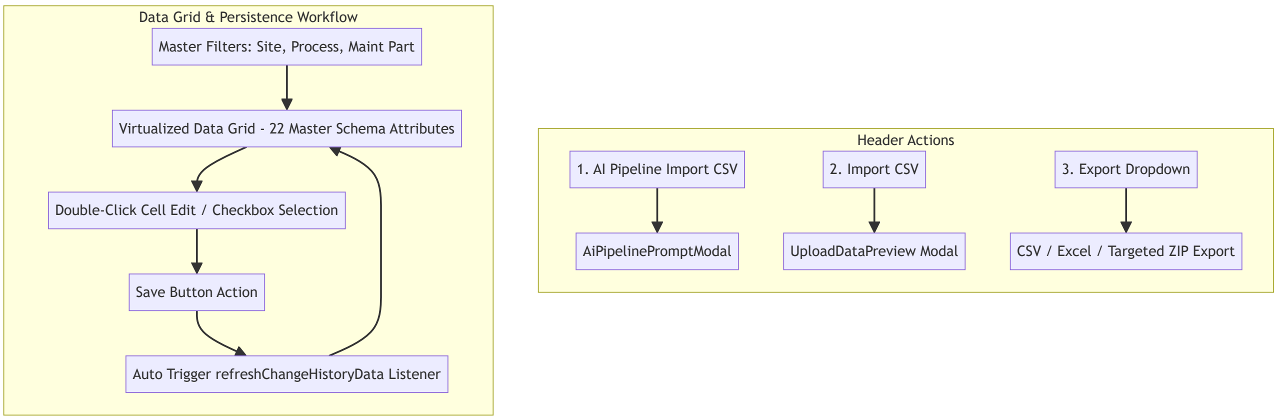
1. Executive Summary & Individual Flow Diagram

1.1 Purpose

The **Change History Data Management Module** serves as the central operational hub for managing, viewing, filtering, inline-editing, batch-deleting, importing, and exporting standardized equipment maintenance records (`ChangeData`).

1.2 Individual Module Flow Diagram

SYSTEM WORKFLOW & PROCESS ARCHITECTURE



2. Functional Requirements

2.1 Header Action Bar Sequence

- **Req-1.1 Action Button Ordering**: The top action bar MUST render buttons in strict left-to-right order:
- **AI Pipeline Import CSV**
- **Import CSV**
- **Export** (Dropdown for CSV, Excel, and Targeted ZIP)
- **Req-1.2 User Identity Tracking**: AI Pipeline import requests MUST capture and attach the authenticated operator's user name.

2.2 Predefined Master Data Schema (22 Attributes)

The module MUST manage records according to the 22 standardized master columns:

`Site`, `Process`, `Maintenance Part`, `Equipment Code`, `Equipment Name`, `W/O Code`, `Report Content`, `BOM`, `Spare Part`, `Work Date`, `Improvement Work`, `Work Purpose`, `Problem Symptom`, `Problem Cause`, `HW Before`, `HW After`, `SW Before`, `SW After`, `Representative Work`, `Priority`, `Category`, `Wotype`.

2.3 Export Functions & Targeted ZIP Package

- **Req-3.1 CSV & Excel Export**: Exports selected table rows (or all filtered rows if none selected) as `.csv` or `.xlsx`.
- **Req-3.2 Targeted ZIP Export**: When the user selects specific rows via table checkboxes, ZIP Export MUST export only the selected record IDs. If no rows are selected, it exports the default record set.

2.4 Automatic Data Refresh

- **Req-4.1 Navigation & Event Triggers**: Navigating to Change History or saving changes in preview modals MUST automatically trigger a data refresh so that table records and dropdown filters reflect the latest state immediately.

3. QC Testing & Acceptance Criteria

- **UI Button Order Test**: Verify that **AI Pipeline Import CSV**, **Import CSV**, and **Export** appear in exact order.
- **Mandatory Field Highlighting Test**: Confirm that only mandatory fields (`Site` , `Process` , `Maintenance Part` , `Equipment Code` , `Equipment Name` , `Representative Work` , `Priority` , `Category`) are highlighted in red when empty.
- **Targeted ZIP Export Test**: Verify that selecting row checkboxes exports a ZIP file containing only those selected records.

Software Requirements Specification (SRS)

AI Pipeline Jobs & Quarantine Modules

Document Control

Item	Details
Document Title	SRS: Jobs Execution Tracking & Quarantine Subsystems
System Modules	EQUAL AI Pipeline Subsystem (`/ai-pipeline/jobs` & `/ai-pipeline/quarantine`)
Document Version	1.0.0 (Pre-Implementation Baseline)
Target Audience	Software Engineers, AI Pipeline Developers, QA / QC Test Engineers

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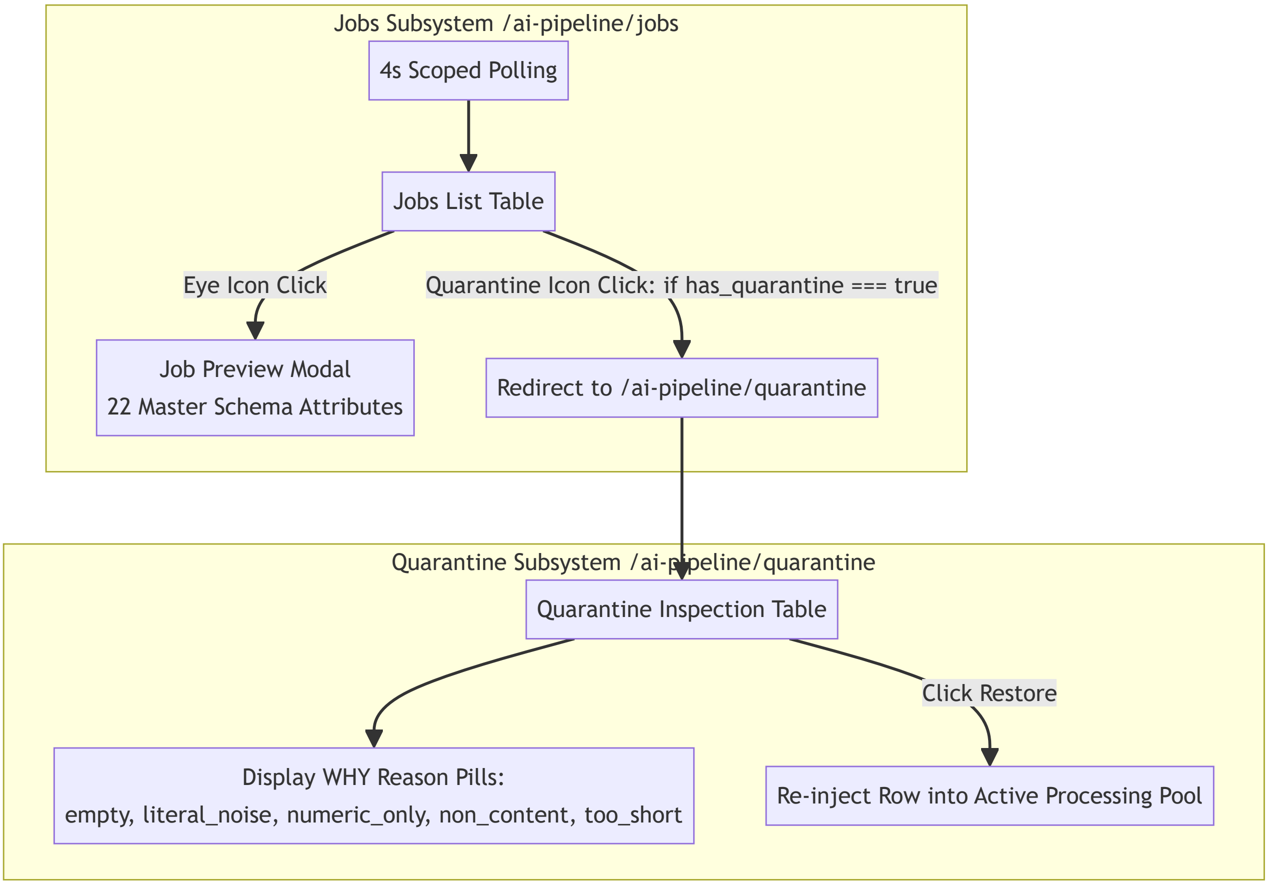
1. Executive Summary & Individual Flow Diagram

1.1 Purpose

The **Jobs Module** and **Quarantine Module** work in tandem to provide asynchronous job tracking, export verification, and isolated row restoration for all multi-format AI data extraction runs.

1.2 Individual Module Flow Diagram

SYSTEM WORKFLOW & PROCESS ARCHITECTURE



2. Jobs Module Specifications (`/ai-pipeline/jobs`)

2.1 Real-Time Job Polling

- **Req-1.1 Polling Scope**: The system MUST poll for job status updates every 4 seconds **only while the Jobs page is active**.
- **Req-1.2 Polling Cleanup**: Navigating away from Jobs MUST stop background polling immediately.

2.2 Table Actions

- **Req-2.1 Eye Icon Preview**: Clicking the Eye icon opens a preview modal rendering extracted rows mapped to the 22 master schema attributes, highlighting missing mandatory fields and enabling inline cell editing.
- **Req-2.2 Conditional Quarantine Icon**: The Quarantine icon button MUST be enabled **only** when `has_quarantine` is `true`. When `has_quarantine` is `false`, the button MUST be disabled and grayed out. Clicking an enabled Quarantine button MUST navigate to `/ai-pipeline/quarantine`.

3. Quarantine Module Specifications (`/ai-pipeline/quarantine`)

3.1 Reasoning Mapping (`WHY` Dictionary)

- Quarantined items MUST display reason explanations mapped from the `WHY` dictionary:
- `empty` : *"The report cell was blank, or held nothing but codes and dates."*
- `literal_noise` : *"The text matched a placeholder phrase (N/A, 없음, 확인중, TBD...)."*
- `numeric_only` : *"The text was only digits."*
- `non_content` : *"The text was only punctuation or symbols."*
- `too_short` : *"Fewer than the minimum characters once codes and dates were removed."*

3.2 Table & Restoration Flow

- **Req-3.1 Table Columns**: `Source` (file name, row number), `W/O code`, `Process · Equipment`, `What the cell held`, `Why` (reason badge + explanation text), and `Action` (Restore button or status).
- **Req-3.2 Restore Action**: Clicking **Restore** puts the row back as a valid report in the active processing pool.

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4. QC Testing & Acceptance Criteria

- **Jobs Polling Lifecycle Test**: Confirm polling starts on entering Jobs page and stops immediately upon leaving.
- **Quarantine Icon Enable/Disable Test**: Verify Quarantine button is enabled only for jobs with `has_quarantine === true`.
- **Quarantine Restore Test**: Confirm clicking **Restore** successfully restores non-blank quarantined rows.

Software Requirements Specification (SRS)

Matrix Inquiry Module

Document Control

Item	Details
Document Title	SRS: Matrix Inquiry & Equipment Heatmap Module
System Module	EQUAL Change Matrix Subsystem (`/change-matrix/matrix-inquiry`)
Document Version	1.0.0 (Pre-Implementation Baseline)
Target Audience	Data Visualization Engineers, Maintenance Managers, QC Test Engineers

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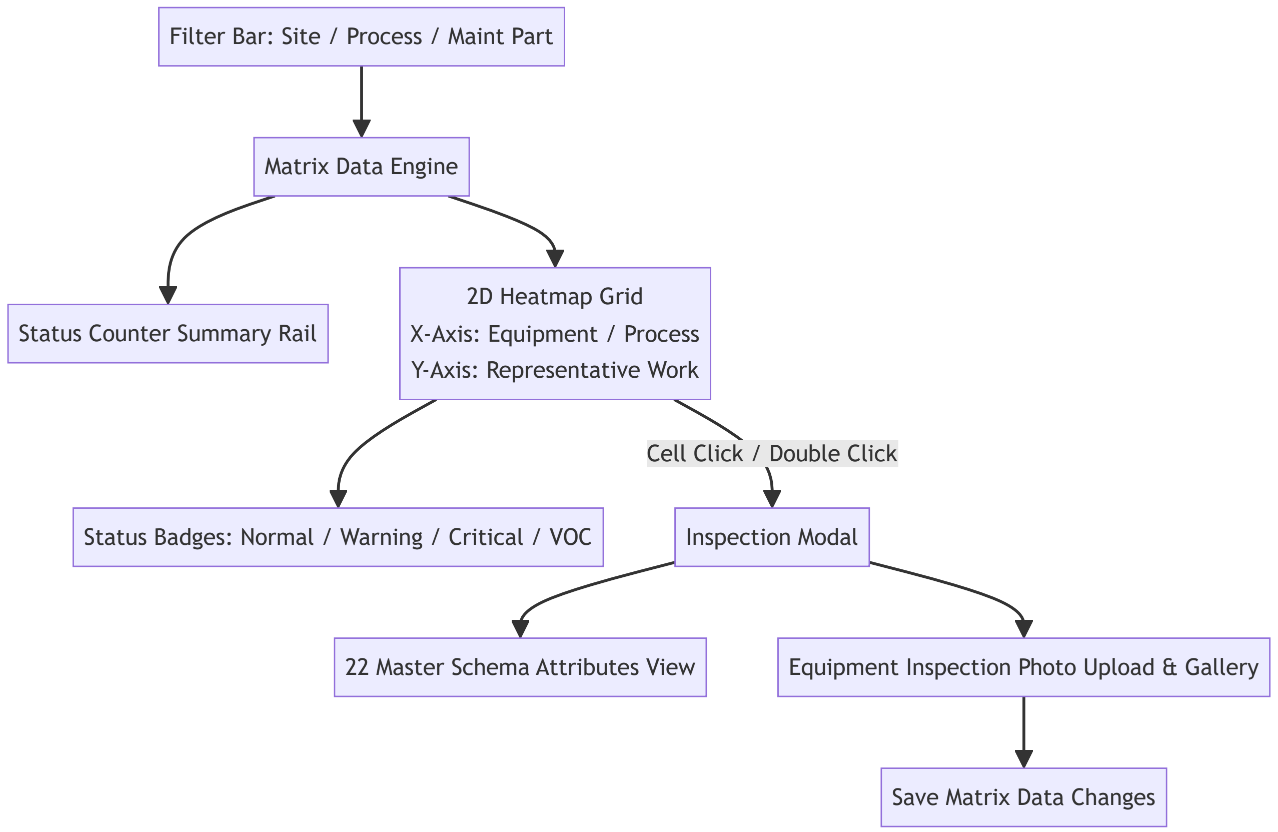
1. Executive Summary & Individual Flow Diagram

1.1 Purpose

The **Matrix Inquiry Module** provides an interactive 2D Heatmap Grid cross-referencing equipment assets against representative work activities, enabling plant engineers to visualize maintenance frequency, operational status, and VOC (Voice of Customer) complaint flags.

1.2 Individual Module Flow Diagram

SYSTEM WORKFLOW & PROCESS ARCHITECTURE



2. Functional Requirements

2.1 2D Heatmap Matrix Grid

- **Req-1.1 Axis Alignment**: X-Axis represents Equipment Items / Process Steps; Y-Axis represents Representative Work Items.
- **Req-1.2 Status Color-Coding**:
- `Normal` : Green badge.
- `Warning / Frequent` : Amber badge.

- `Critical / VOC Flagged` : Red badge.

2.2 Inspection Modal & Media Attachments

- **Req-2.1 Detail Inspection**: Clicking a matrix cell opens a modal displaying detailed work order records across the 22 master schema fields.
- **Req-2.2 Photo Attachments**: Supports uploading, viewing, and managing equipment inspection photos.

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3. QC Testing & Acceptance Criteria

- **Filter Synchronization**: Grid MUST reload dynamically when Site or Process filter selections change.
- **Heatmap Color Accuracy**: Status color badges MUST accurately reflect equipment criticality and VOC status.
- **Photo Upload Verification**: Attached inspection photos MUST render correctly within the cell detail modal.

Software Requirements Specification (SRS)

Maintenance Plan (MP) List Inquiry Module

Document Control

Item	Details
Document Title	SRS: Maintenance Plan (MP) List Inquiry Module
System Module	EQUAL Change Matrix Subsystem (`/change-matrix/mp-list-inquiry`)
Document Version	1.0.0 (Pre-Implementation Baseline)
Target Audience	Software Engineers, Maintenance Leads, QC Auditors

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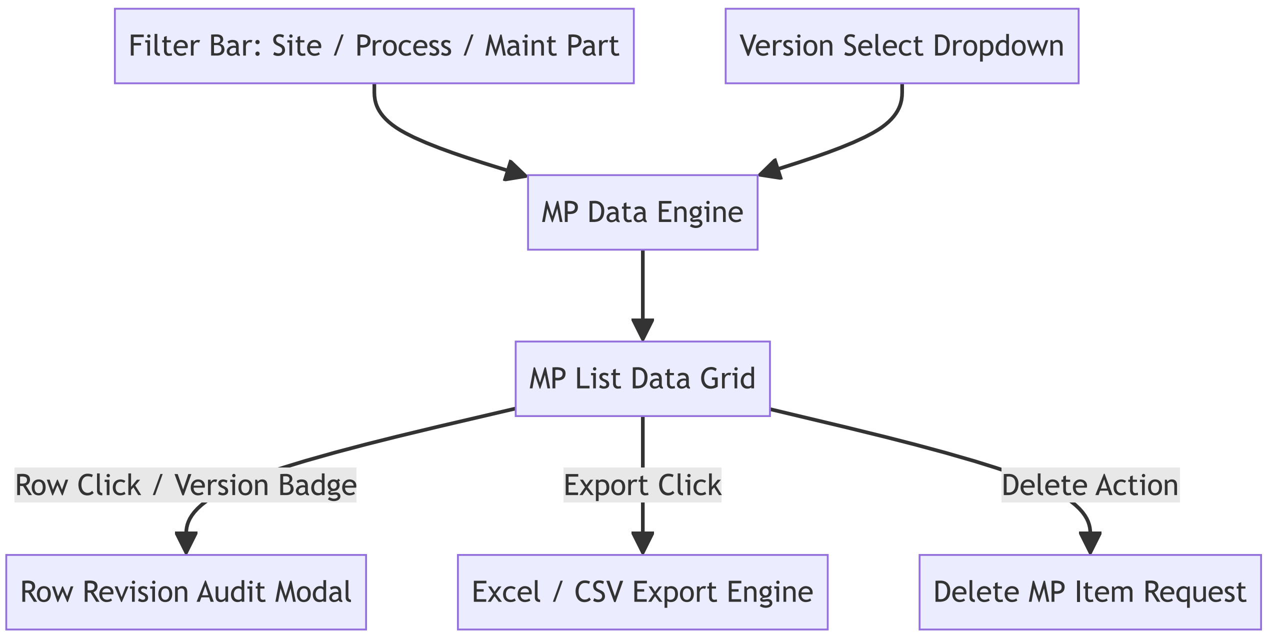
1. Executive Summary & Individual Flow Diagram

1.1 Purpose

The **MP List Inquiry Module** provides maintenance engineers and auditors with a comprehensive interface to query, filter, compare versions, and audit revision histories of Maintenance Plan (MP) records across manufacturing facilities.

1.2 Individual Module Flow Diagram

SYSTEM WORKFLOW & PROCESS ARCHITECTURE



2. Functional Requirements

2.1 Multi-Criteria Data Grid

- **Req-1.1 Filtering**: Dynamically filter MP items by Site, Process, Maintenance Part, and active Version.
- **Req-1.2 Master Attributes**: Display Equipment Code, Equipment Name, Representative Work, Version Numbers, and Active Status.

2.2 Version Management & Revision Audit

- **Req-2.1 Version Switching**: Support switching active MP versions from a dropdown menu.
- **Req-2.2 Row Revision Audit**: Clicking a row's version badge opens a historical audit modal displaying past modifications to that specific MP item.
- **Req-2.3 Delete MP Item**: Support removing individual MP records with confirmation.

3. QC Testing & Acceptance Criteria

- **Version Switch Test**: Changing active version MUST update all grid rows immediately.
- **Revision History Test**: Opening row revision history MUST display all historical changes in chronological order.

- **Deletion Audit Test**: Deleting an item MUST remove it from the active view.

Software Requirements Specification (SRS)

Maintenance Plan (MP) List Management Module

Document Control

Item	Details
Document Title	SRS: MP List & Specification Data Management Workspace
System Module	EQUAL Change Matrix Subsystem (`/change-matrix/mp-list-management`)
Document Version	1.0.0 (Pre-Implementation Baseline)
Target Audience	System Administrators, Specification Engineers, QC Test Engineers

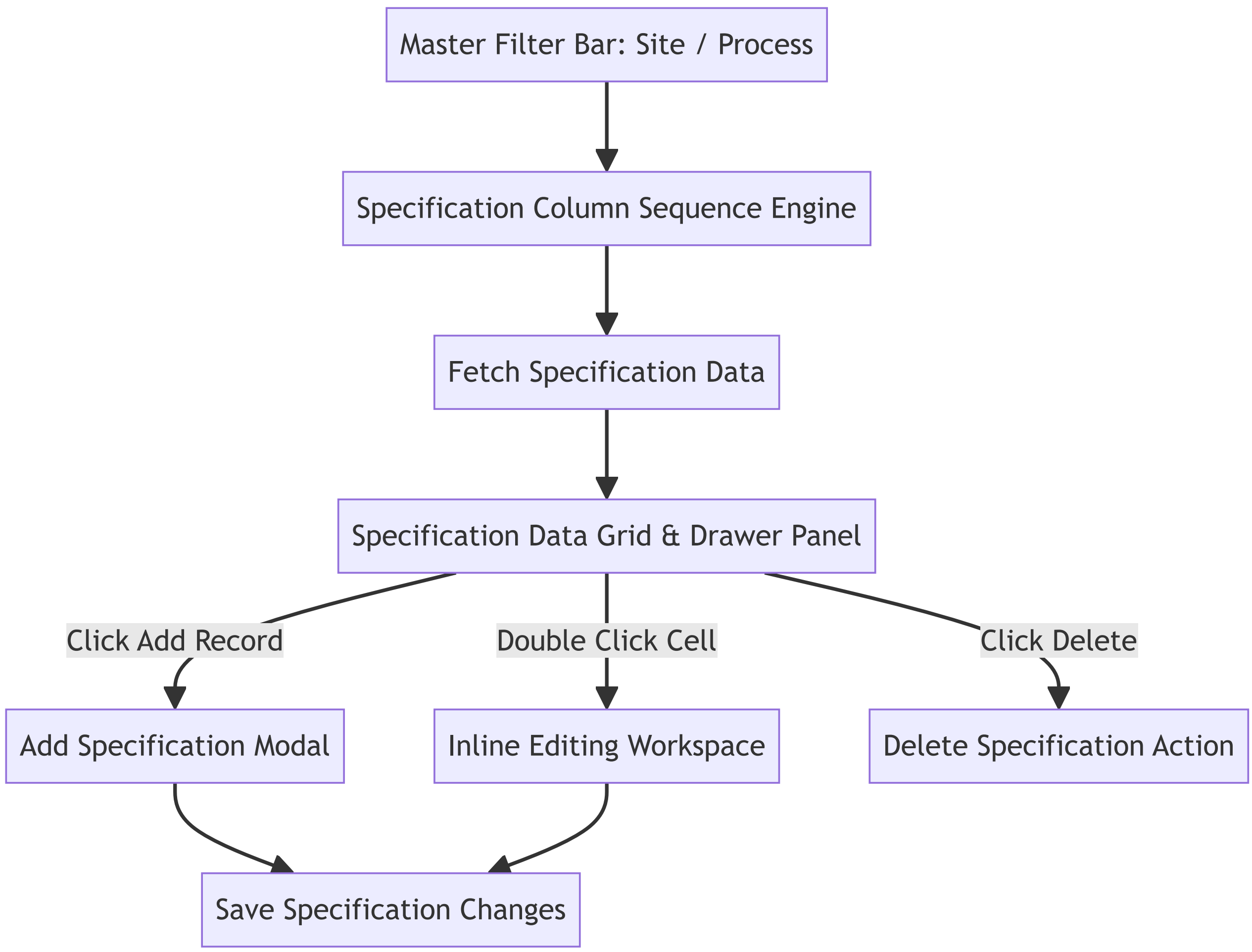
1. Executive Summary & Individual Flow Diagram

1.1 Purpose

The **MP List Management Module** serves as the primary authoring workspace for configuring, editing, reordering, adding, deleting, and persisting Maintenance Plan (MP) and Specification Data records (`SpecData`).

1.2 Individual Module Flow Diagram

SYSTEM WORKFLOW & PROCESS ARCHITECTURE



2. Functional Requirements

2.1 Specification Master Column Ordering Engine

- **Req-1.1 Master Column Definitions**: Loads specification master column definitions.
- **Req-1.2 Sequence Ordering**: Grid columns MUST be ordered strictly by `sequence`, guaranteeing uniform attribute order across all plant views.

2.2 Authoring & Workspace Management

- **Req-2.1 Add Record**: Provides an **Add Record** modal allowing operators to define new equipment parameters.
- **Req-2.2 Inline & Drawer Editing**: Supports double-click inline cell editing and drawer panel editing.
- **Req-2.3 Delete Record**: Supports deleting individual specification items with confirmation.

3. QC Testing & Acceptance Criteria

- **Column Sequence Test**: Verify that grid columns appear in exact order defined by column sequence numbers.
- **Add & Save Test**: Adding a specification record MUST immediately update the workspace grid.
- **Delete Test**: Deleting a record MUST remove it from the active dataset.

Software Requirements Specification (SRS)

Multi-Format File Processing System

Document Control

Item	Details
Document Title	SRS: Multi-Format File Ingestion & Document Processing System
Supported File Formats	PPT/PPTX, Excel (XLSX/XLS/CSV), PDF, Images (PNG/JPG), EML Emails
Primary AI Inference Engine	GPT Opus 120B Large Language Model
Document Version	1.0.0 (Pre-Implementation Baseline)
Target Audience	Systems Engineers, Integration Architects, QA Testers

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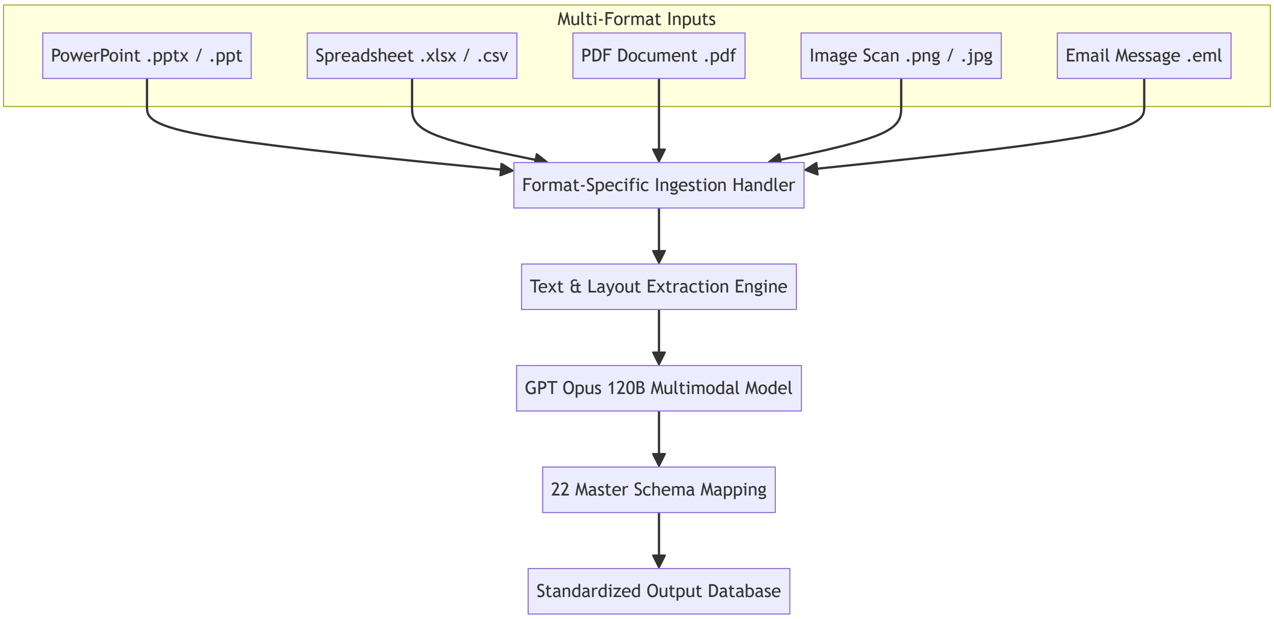
1. Functional Scope & Individual Flow Diagram

1.1 Objective

The Multi-Format File Processing System provides a unified ingestion framework capable of extracting operational maintenance records from multi-modal source documents.

1.2 Individual Module Flow Diagram

SYSTEM WORKFLOW & PROCESS ARCHITECTURE



2. Ingestion Specifications by File Format

2.1 PowerPoint Presentation Decks (PPT / PPTX)

- **Requirements**:
- Extract text contents from slides, text boxes, callout shapes, and embedded tables.
- Parse slide titles to contextualize maintenance activity locations.

2.2 Spreadsheets (XLSX / XLS / CSV)

- **Requirements**:
- Auto-detect table boundaries, multi-header rows, and merged cells.
- Parse multiple worksheets within a single workbook.

2.3 PDF Inspection Reports (PDF)

- **Requirements**:
- Process both native digital PDFs and scanned image-based PDFs.
- Retain tabular layout structures during text extraction.

2.4 Diagnostic Images & Photos (PNG / JPG)

- **Requirements**:
- Extract text embedded within maintenance screenshots, equipment nameplates, and meter readings.
- Feed visual context directly to the multimodal AI processing pipeline.

2.5 Email Communications (EML)

- **Requirements**:
- Extract email headers (Sender, Date, Subject) and body text.
- Process attached documents (e.g., attached spreadsheets or PDF reports).

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3. QC Testing & Acceptance Criteria

- **Format Resilience Test**: The system MUST process all 5 supported file types in a single execution queue without failing.
- **Data Completeness Test**: Extracted fields from all file types MUST map into the 22 master schema attributes.
- **Error Handling**: Corrupt or unreadable files MUST generate a clear error message and isolate affected items into Quarantine.

Software Requirements Specification (SRS)

Non-Standard Format Data Extraction System & Verification Engine

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Document Control

Item	Details
Document Title	SRS: System for Extracting Data from Non-Standard Formats & Performance Validation
System Module	Data Extraction & Verification Engine
Primary AI Engine	GPT Opus 120B Large Language Model
Document Version	1.0.0 (Pre-Implementation Baseline)
Target Audience	Software Architects, QA Lead Engineers, Data QC Testers

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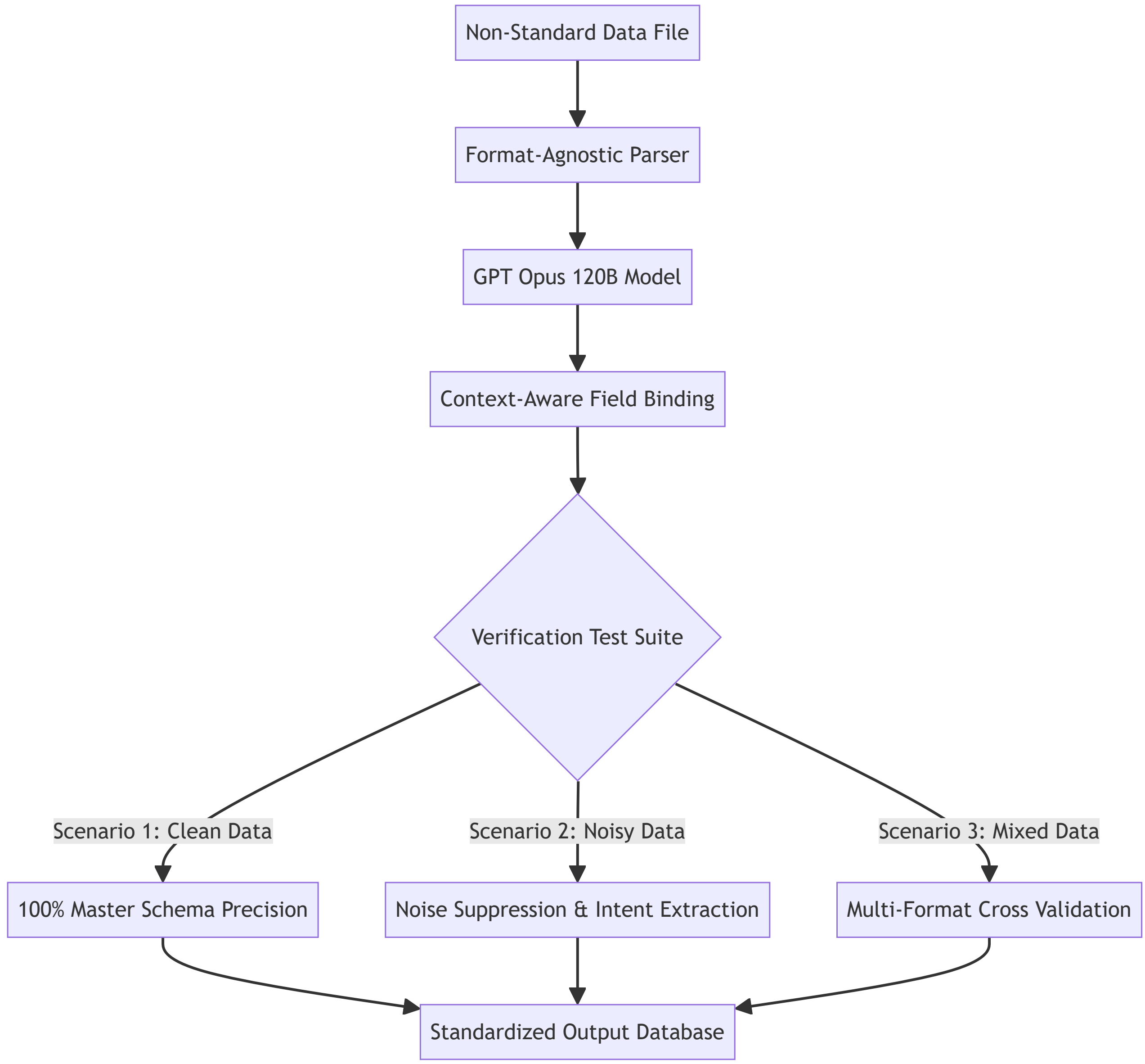
1. Executive Summary & Individual Flow Diagram

1.1 Purpose

Legacy plant maintenance logs are frequently stored in non-standard formats with irregular layout patterns, merged spreadsheet cells, informal column titles, and varying terminology. This specification defines the requirements for a resilient extraction system capable of interpreting non-standard inputs, normalizing raw content into standardized schemas, and validating extraction accuracy using defined verification test suites.

1.2 Individual Module Flow Diagram

SYSTEM WORKFLOW & PROCESS ARCHITECTURE



2. Theoretical Processing Logic

- **Format Agnostic Parsing**: The system parses input documents regardless of layout structure or custom column names.
- **Context-Aware Field Binding**: Using deep semantic understanding provided by **GPT Opus 120B**, non-standard column headers (e.g., "설비명칭", "Machine ID", "작업내용") are automatically mapped to standard target fields (`Equipment Name` , `Equipment Code` , `Report Content`).
- **Validation & Isolation**: Entries failing basic mandatory constraints are highlighted for user review or isolated in Quarantine.

3. Quality Control (QC) Verification Test Scenarios

Quality Control engineers MUST evaluate system performance across three specific verification scenarios:

3.1 Verification Scenario 1: Clean Data Benchmark

- **Input Specification**: Standardized files with clear column headers and complete entries.
- **Expected QC Outcome**:
- Zero missing mandatory fields (`Site` , `Process` , `Maintenance Part` , `Equipment Code` , `Equipment Name` , `Representative Work` , `Priority` , `Category`).
- 100% correct data type enforcement (e.g., ISO date formatting).

3.2 Verification Scenario 2: Noisy Data Benchmark

- **Input Specification**: Files containing OCR noise, placeholder tokens ("N/A", "확인중", "TBD"), text truncations, and informal slang.
- **Expected QC Outcome**:
- Automatic suppression of meaningless noise tokens.
- Accurate extraction of real underlying maintenance intent.
- Flagging incomplete records without causing processing failures.

3.3 Verification Scenario 3: Mixed Data Benchmark

- **Input Specification**: Batch uploads combining spreadsheets, presentation decks, PDF invoices, image scans, and email messages.
- **Expected QC Outcome**:
- Unified output dataset standardized to the master target schema.
- Zero data loss or corruption during multi-format batch ingestion.

4. Operational Requirements

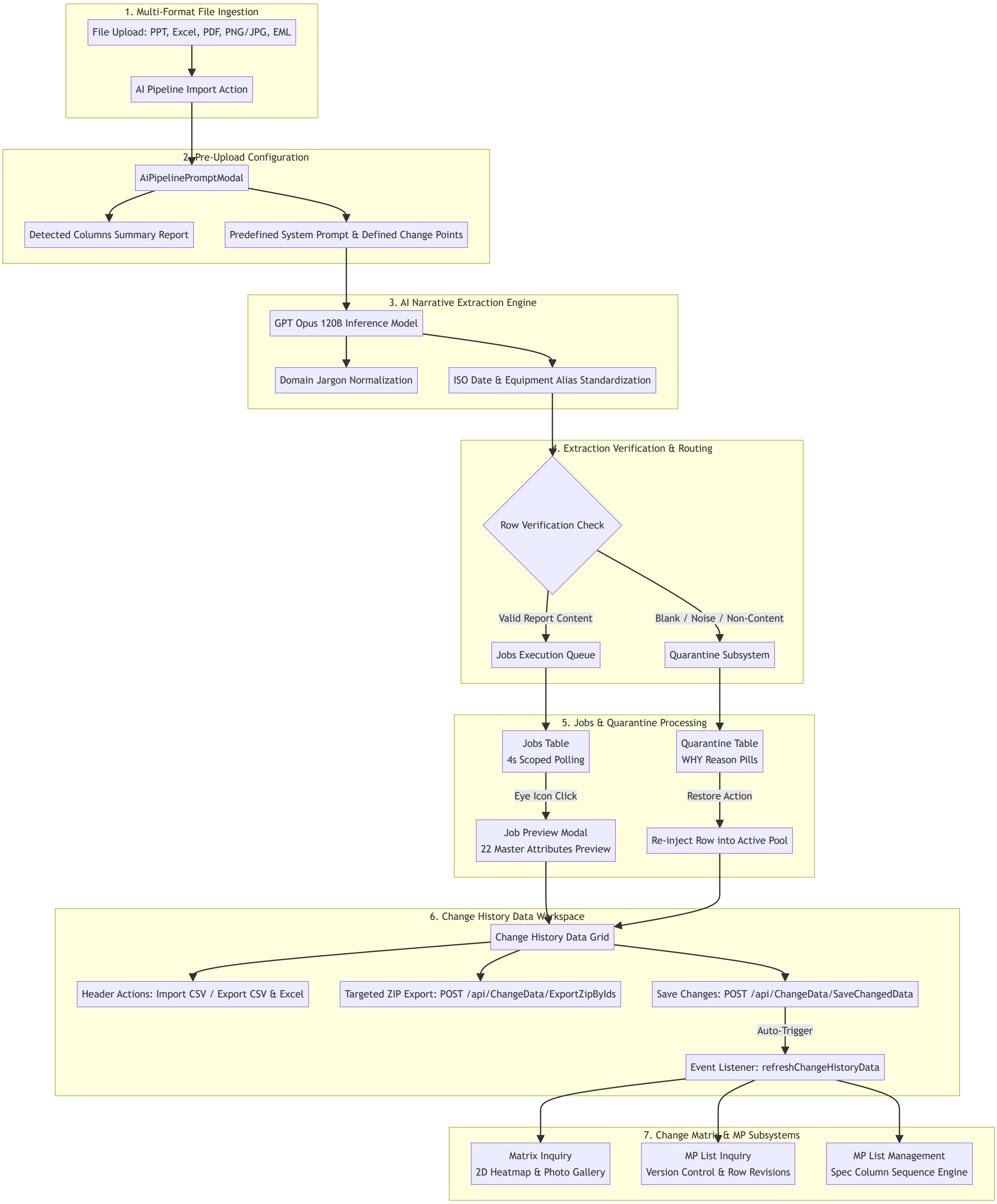
- **Accuracy Threshold**: Minimum 95% field extraction accuracy on noisy data inputs.
- **Fault Tolerance**: Malformed rows MUST be quarantined without breaking the pipeline.
- **Auditability**: All extraction actions MUST retain line-level source traceability.

System Specification: End-to-End Flow Diagram & Data Lifecycle

1. System Architecture & Workflow Diagram

This document illustrates the complete end-to-end operational flow of the EQUAL system—from initial multi-format file upload through AI narrative extraction, job monitoring, quarantine handling, master schema standardization, change history management, matrix heatmap visualization, and maintenance plan (MP) versioning.

SYSTEM WORKFLOW & PROCESS ARCHITECTURE



2. Stage-by-Stage Operational Summary for QC Testing

Stage #	Stage Name	System Action	QC Verification Expectation
1	**Multi-Format Ingestion**	Ingests `.xlsx`, `.csv`, `.pdf`, `.pptx`, `.png`, `.eml` files.	System accepts all 5 supported file types without rejection.
2	**Pre-Upload Report**	Displays column headers and editable prompt with Change Points.	Column summary tags match input file headers exactly.
3	**AI Extraction**	GPT Opus 120B maps text to 22 master schema columns.	Jargon normalized; dates converted to ISO-8601 (`YYYY-MM-DD`).
4	**Routing & Isolation**	Valid rows sent to Jobs; blank/noise sent to Quarantine.	Invalid/empty rows isolated without breaking the job run.
5	**Jobs & Quarantine**	4s polling on Jobs; Restore action on Quarantine.	Polling active only on Jobs page; Restore returns row to pool.
6	**Change History Data**	22-attribute grid editing, targeted ZIP export by IDs.	Selected checkbox IDs exported in ZIP package.
7	**Matrix & MP Modules**	Heatmap status badges, MP versioning, column sequence.	Column order strictly matches sequence numbers.

Software Requirements Specification (SRS)

Unstructured & Semi-Structured Data Extraction Pipeline

Document Control

Item	Details
Document Title	SRS: Unstructured & Semi-Structured Data Ingestion and Schema Standardization
Target Subsystem	EQUAL Data Extraction & Schema Standardization Pipeline
Core Inference Engine	GPT Opus 120B Large Language Model
Document Version	1.0.0 (Pre-Implementation Baseline)
Primary Audience	Product Managers, Data Engineers, QC / QA Test Engineers, Domain Engineers

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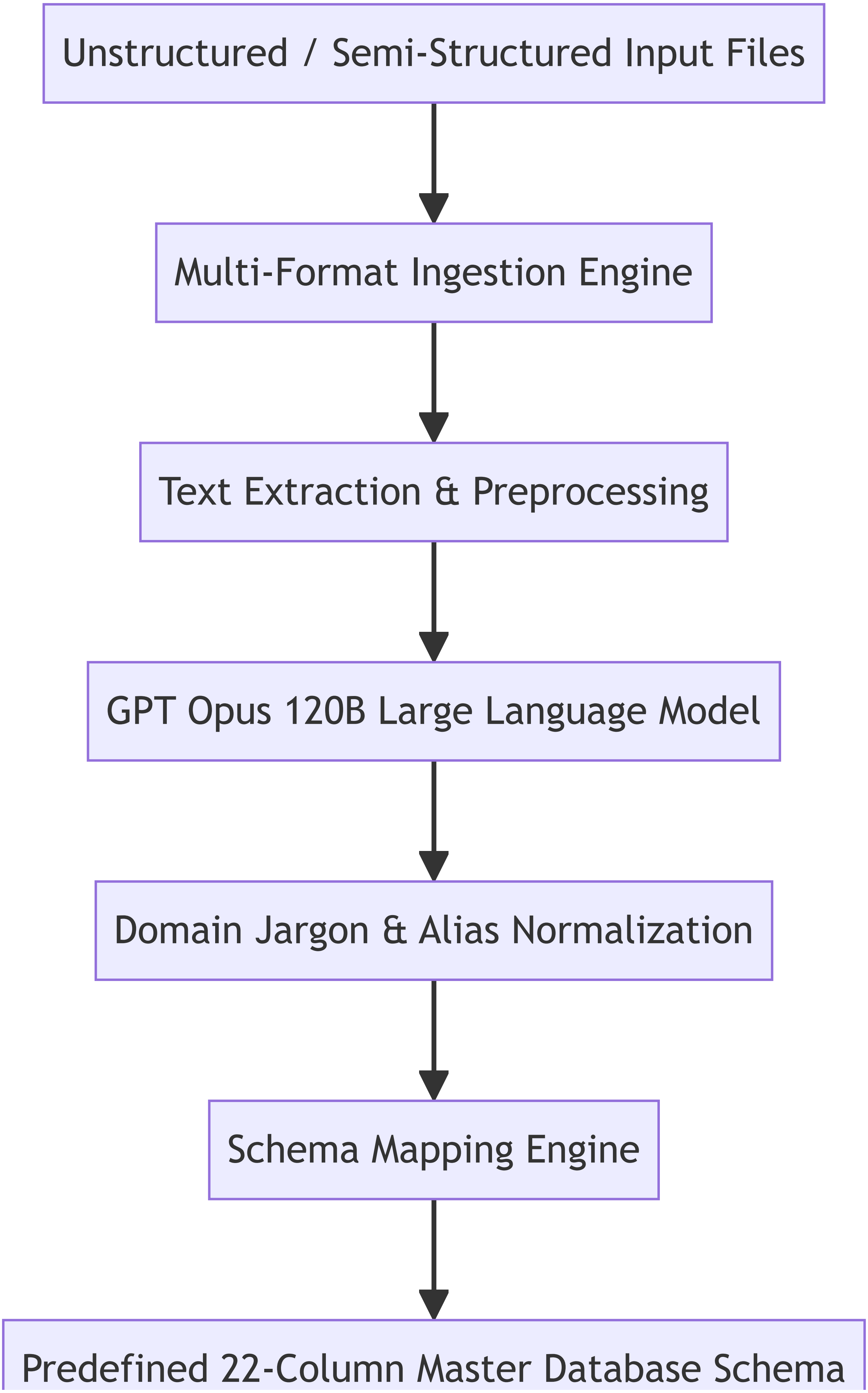
1. Executive Summary & Project Purpose

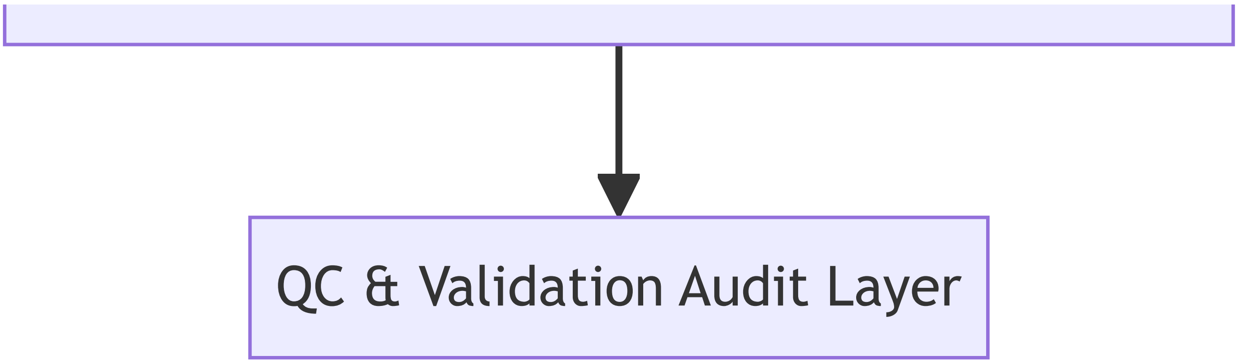
1.1 Purpose

Industrial maintenance operations generate vast amounts of unstructured and semi-structured documentation —such as free-text work logs, technical inspection reports, equipment audit slide decks, email communications, and scanned diagnostic invoices. These documents contain critical insights regarding machine health, failure causes, and maintenance history, but lack standardized structure.

This System Requirements Specification (SRS) defines the functional, data, processing, and Quality Control (QC) requirements for an AI-powered data extraction pipeline. Powered by the **GPT Opus 120B** model, the system ingests non-standard multi-format operational documents, interprets domain-specific maintenance narratives, normalizes equipment terminology, and maps extracted attributes into a **predefined 22-column master database schema**.

SYSTEM WORKFLOW & PROCESS ARCHITECTURE


Unstructured / Semi-Structured Input Files
Multi-Format Ingestion Engine
Text Extraction & Preprocessing
GPT Opus 120B Large Language Model
Domain Jargon & Alias Normalization
Schema Mapping Engine
Predefined 22-Column Master Database Schema



2. Predefined Master Database Schema (22 Attributes)

The extraction engine MUST convert and normalize arbitrary input fields into the following 22 standardized master target attributes:

Column #	Standardized Attribute	Data Type	Mandatory Rule	Description & QC Validation Expectation
1	`Site`	String	Mandatory	Manufacturing plant or facility location (e.g., "Suwon Site A1").
2	`Process`	String	Mandatory	Production line or process step (e.g., "03. Molding / Coating").
3	`Maintenance Part`	String	Mandatory	Maintenance group or equipment class (e.g., "0303. R2 Coater").
4	`Equipment Code`	String	Mandatory	Unique hardware asset tag / code (e.g., "H0303001").
5	`Equipment Name`	String	Mandatory	Descriptive equipment label (e.g., "Coater(3R2)_450").
6	`W/O Code`	String	Optional	Work Order or Work Instruction identifier code (e.g., "W004512547").
7	`Report Content`	String	Optional	Full narrative of maintenance activity or technician observation.
8	`BOM`	String	Optional	Bill of Materials reference or part catalog identifier.
9	`Spare Part`	String	Optional	List of replaced components or spare parts utilized during repair.
10	`Work Date`	ISO Date/Time	Optional	Timestamp or date when maintenance was performed (YYYY-MM-DD format).

Column #	Standardized Attribute	Data Type	Mandatory Rule	Description & QC Validation Expectation
11	`Improvement Work`	String	Optional	Concise summary of corrective action taken.
12	`Work Purpose`	String	Optional	Stated objective or justification for the maintenance intervention.
13	`Problem Symptom`	String	Optional	Observed machine anomaly, failure symptom, or operational warning.
14	`Problem Cause`	String	Optional	Identified root cause of the equipment breakdown.
15	`HW Before`	String	Optional	Hardware state, model, or count prior to maintenance.
16	`HW After`	String	Optional	Hardware state, model, or count following maintenance.
17	`SW Before`	String	Optional	Software version or firmware parameter prior to change.
18	`SW After`	String	Optional	Software version or firmware parameter following change.
19	`Representative Work`	String	Mandatory	Standardized cluster title for the maintenance activity.
20	`Priority`	Enum	Mandatory	Criticality level ("Urgent", "Important", "Normal").
21	`Category`	Enum	Mandatory	Maintenance classification ("Quality", "Safety", "Productivity", "Other").
22	`Wotype`	Enum	Optional	Work order classification type (e.g., "CM", "BM", "PM").

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3. Functional Requirements & System Capabilities

3.1 AI Narrative Interpretation (GPT Opus 120B)

- **FR-01.1 Contextual Interpretation**: The system SHALL process unstructured technical Korean and English prose to accurately discern root causes from symptoms and corrective actions from work purposes.
- **FR-01.2 Domain Jargon Normalization**: The AI engine SHALL automatically translate colloquial plant jargon, abbreviations, and informal equipment aliases into official master terms.

- **FR-01.3 Date & Numeric Standardization**: All extracted dates SHALL be converted to standard ISO-8601 (`YYYY-MM-DDTHH:mm:ss`) format regardless of source input formatting (e.g., "2019/4/24", "19.04.24", "Apr 24 2019").

3.2 Pre-Upload Change Point Prompt Configuration

- **FR-02.1 Change Point Customization**: Operators SHALL be able to review and adjust AI extraction directives and prompt parameters prior to starting pipeline ingestion.
- **FR-02.2 Real-Time Schema Preview**: The system SHALL provide a preliminary summary report of detected input columns before executing full model extraction.

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4. QC Testing & Quality Assurance Verification Plan

Quality Control (QC) engineers SHALL verify the extraction accuracy and system resilience using three structured test suites:

4.1 Verification Scenario A: Clean Data Test Suite

- **Dataset**: Standard structured spreadsheets with complete headers and well-formatted text.
- **Pass Criteria**:
- 100% field extraction precision across all 22 master schema columns.
- Zero missing mandatory fields (`Site`, `Process`, `Maintenance Part`, `Equipment Code`, `Equipment Name`, `Representative Work`, `Priority`, `Category`).

4.2 Verification Scenario B: Noisy Data Test Suite

- **Dataset**: Free-text logs containing OCR artifacts, Korean/English typos, missing punctuation, placeholder phrases ("N/A", "TBD", "확인중"), and slang.
- **Pass Criteria**:
- System successfully filters out noise phrases and flags invalid rows for Quarantine.
- Correct identification of core maintenance intent despite spelling errors.

4.3 Verification Scenario C: Mixed Format Test Suite

- **Dataset**: Multi-page PDFs containing embedded tables, PPTX slides, scanned inspection images, and email attachment threads.
- **Pass Criteria**:
- Successful end-to-end extraction across all document formats without memory leak or process crash.
- Consistent schema mapping to the 22 master target fields across all file types.

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5. Non-Functional Requirements (NFR)

- **NFR-01 Extraction Speed**: Total processing time SHALL not exceed 3.0 seconds per record for standard multi-row files.
- **NFR-02 Model Model Integrity**: The pipeline SHALL utilize the **GPT Opus 120B** model to guarantee high-reasoning accuracy on domain-specific maintenance queries.
- **NFR-03 Reliability**: The system SHALL handle corrupt file inputs gracefully by sending non-parsable records to a Quarantine view without halting the overall job execution.
